Analyzing Bias Propogation in Recommender Feedback Loops Using Embeddings

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Project Guide

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Feedback Loop Bias Example

Spotify

Listen to *Popular Artist* → Recommend Similar Popular Songs → User Streams Recommended Songs → Model Learns Strong Preference for Popular Artists → Recommend More Popular Music

Feedback Loop: Recommendations shape user listening behavior, which becomes new training data. Repeated cycles reinforce popularity bias and reduce exposure to less-popular artists and diverse genres.

Introduction

- Recommender systems represent users and items as **latent vector embeddings**.
- Recommendations influence future user interactions, creating a **closed feedback loop**.
- Repeated retraining can amplify **popularity bias** and cause changes in embedding geometry.

Domain of Interest

- Recommender Systems
- Vector Embeddings
- Closed Feedback Loops
- Embedding Geometry

Problem Context

- **Systemic Observational Bias**: Historical training logs contain inherent observational bias and do not accurately reflect true user tastes.
- **Algorithmic Reinforcement**: Standard collaborative models disproportionately favor popular items while neglecting the majority of catalog items.
- **Dynamic Feedback Amplification**: Rather than staying static, feedback loop bias is self-reinforcing, growing progressively stronger with each subsequent retraining generation.
- **Degradation of Personalization**: Niche users are increasingly deprived of recommendations that cater to their genuine, diverse interests.

Literature Survey

Paper: *Dynamic Feedback Loops in Recommender Systems*
Authors: Zoralioglu & Yalcin (2026)

Significance to Our Project

- Provides baseline conceptual framework for multi-generation ($T = 10$) closed-loop simulations.
- Demonstrates progressive exposure degradation and calibration loss across retraining iterations.

Research Gap Identified

- Evaluates system degradation strictly via surface-level recommendation lists.
- Lacks vector-space geometric tracking (fails to measure embedding drift velocity, coordinate variance collapse, or spatial deformation).

Literature Survey

Paper: *Feedback Loop and Bias Amplification in Recommender Systems*
Authors: Mansoury et al. (2020)

Significance to Our Project

- Pioneers empirical closed-loop offline simulation across standard CF algorithms (`UserKNN`, `BPR`, `MostPopular`).
- Proves feedback loops amplify popularity bias (Matthew Effect) and systematically reduce catalog diversity.

Research Gap Identified

- Relies on static, non-adaptive user acceptance rules without position decay or popularity conformity dynamics.
- Fails to analyze internal representation dynamics (cannot differentiate pointwise origin collapse from pairwise periphery dispersion).

Literature Survey

Paper: *Cornac: A Comparative Multimodal Framework for Recommendation*
Authors: Salah et al. (2020)

Significance to Our Project

- Core recommendation execution engine utilized in our research framework (Phase 2 benchmarking & Subsystem 2 adapters).
- Enables unified, reproducible training of matrix factorization (`MF`, `PMF`, `BPR`) and memory-based models.

Research Gap Identified

- Architected exclusively for static, single-pass offline model evaluation.
- Lacks native support for multi-generation feedback loop execution, dynamic data buffer accumulation, and geometric coordinate drift tracking.

Literature Survey

Paper: *Causal Disentanglement for Tackling Popularity Bias (CDSRec)*
Authors: Liu et al. (2026)

Significance to Our Project

- Establishes a Structural Causal Model (SCM) to separate intrinsic item quality from trend exposure bias.
- Serves as the state-of-the-art causal debiasing reference benchmark for Phase 4 mitigation strategies.

Research Gap Identified

- High architectural complexity (Markov chains, DAGs, VAEs) tailored for single-pass sequential interaction logs.
- Does not measure geometric embedding space contraction or latent coordinate drift under continuous closed-loop retraining.

Literature Survey

Paper: *How Algorithmic Confounding Increases Homogeneity and Decreases Utility*
Authors: Chaney et al. (2018)

Significance to Our Project

- Early multi-generation agent simulation showing recommendation loops cause user preference homogenization and utility decay.
- Demonstrates how user interaction choices become confounded by prior algorithmic exposure.

Research Gap Identified

- Relies on idealized synthetic user utility functions rather than empirical interaction datasets (e.g., MovieLens 1M).
- Ignores latent representation mechanics, vector coordinate space dynamics, and model-specific geometric failure modes.

Baseline & Survey Literature Summary

Paper	Focus / Methodology	Key Findings	Limitations / Gaps
Chen et al. (2023)	Debiasing survey (IPS, Causal, RL).	Identifies feedback loops as primary bias driver.	Qualitative survey; lacks empirical vector dynamics.
Ma et al. (2019)	MacridVAE via Variational Autoencoders.	Separates macro/micro intentions in latent space.	Learned dimensions lack interpretability without attributes.
Kang & McAuley (2018)	SASRec (Self-Attention Sequential).	Captures long/short-term sequential preferences.	Cannot scale efficiently; ignores feedback loop vector drift.
Hidasi et al. (2016)	GRU4Rec (Gated Recurrent Units).	Models session click transitions.	Relies strictly on item IDs; ignores content features.
Rendle et al. (2009)	BPR (Pairwise SGD ranking).	Optimizes relative item ordering over implicit logs.	Static model; vulnerable to periphery dispersion under loops.

Critical Research Gaps

- **Research Gap 1: Static vs. Dynamic Bias**
  - Existing mitigation strategies treat bias predominantly as a static problem.
  - Current models fail to account for how sequential feedback loops compound and amplify bias across iterative retraining cycles.
- **Research Gap 2: Structural Vector Analysis**
  - Prior literature emphasizes surface recommendation metrics over internal latent representations.
  - Standard benchmarks do not quantify the geometric drift, topological deformation, or contraction of vector embeddings during retraining loops.

Bridging the Research Gap

- **Mathematical Drift Metrics**
  - Formulate quantitative metrics to track embedding displacement vectors, trajectory divergence, and contraction velocity across retraining iterations.
- **Latent Space Preservation**
  - Systematically design and evaluate algorithmic safeguards, regularization constraints, and update bounds that prevent latent space collapse.
- **Catalog Coverage & Fairness**
  - Preserve long-term item catalog exposure equity and safeguard recommendation quality for user groups with niche preferences.

Problem Statement

Problem

Analyze and mitigate dynamic bias propagation in closed-loop recommender systems by tracking and stabilizing latent embedding drift and spatial contraction across retraining cycles to prevent catalog coverage collapse and protect niche user preferences.

Project Objectives

Primary Objective

Analyze and mitigate dynamic bias propagation in closed-loop recommender systems by tracking geometric changes in vector embedding space.

Specific Objectives

- Build Simulation Framework: Construct a closed-loop simulation environment to model multi-generation recommendation cycles.
- Quantify Vector Contraction: Calculate the velocity of embedding space contraction to measure diversity loss and popularity concentration.
- Evaluate Mitigation Methods: Test alternative recommendation algorithms and bias mitigation techniques to stabilize vector trajectories.

Proposed Solution Methodology – Overview

- The system consists of five sequential stages: **baseline anchoring**, **model recommendation**, **stochastic click simulation**, **data ingestion (the feedback loop)**, and **diagnostic auditing**.

- Two core pipelines define the system: a **closed-loop simulation pipeline** (modeling user clicks based on exposure, relevance, and conformity) and a **4-tier diagnostic pipeline** (measuring popularity bias propagation and latent space degradation).

System Setup & Baseline Initialization

- **Data Ingestion:** Ingest the MovieLens-1M dataset and binarize the historical user-item interactions.
- **Baseline Training:** Train the suite of recommender models (Matrix Factorization, BPR, PMF, UserKNN) on this initial dataset.
- **Preference Freezing:** Extract and freeze a "Ground-Truth Anchor" from these baseline models. This serves as a permanent, uncorrupted record of the users' original preferences before the simulation begins.

Simulating User Interactions

- **Recommendation Generation:** Generate Top-K candidate lists for all users from freshly trained models.
- **Exposure Discount:** Scale click likelihood via a position-based rank decay.
- **Relevance Matching:** Score true intrinsic utility against the Frozen Ground-Truth Anchor.
- **Social Conformity:** Inject a bandwagon boost for platform-wide popular items.
- **Click Sampling:** Synthesize exposure, relevance, and conformity into final clicks under session fatigue caps.

Closing the Loop & Measuring Bias

- **Feedback Loop Ingestion:** Append simulated clicks into the interaction matrix as new observations.
- **Iterative Retraining:** Retrain models on augmented logs to simulate self-reinforcing bias cycles.
- **Macro-Distribution Analysis:** Track catalog concentration via Exposure Gini and Catalog Coverage.
- **Subgroup Taste Distortion:** Measure drift between niche user recommendations and baseline profiles.
- **Latent Degradation:** Track representation collapse via Effective Spectral Rank on aligned embeddings.

System Architecture

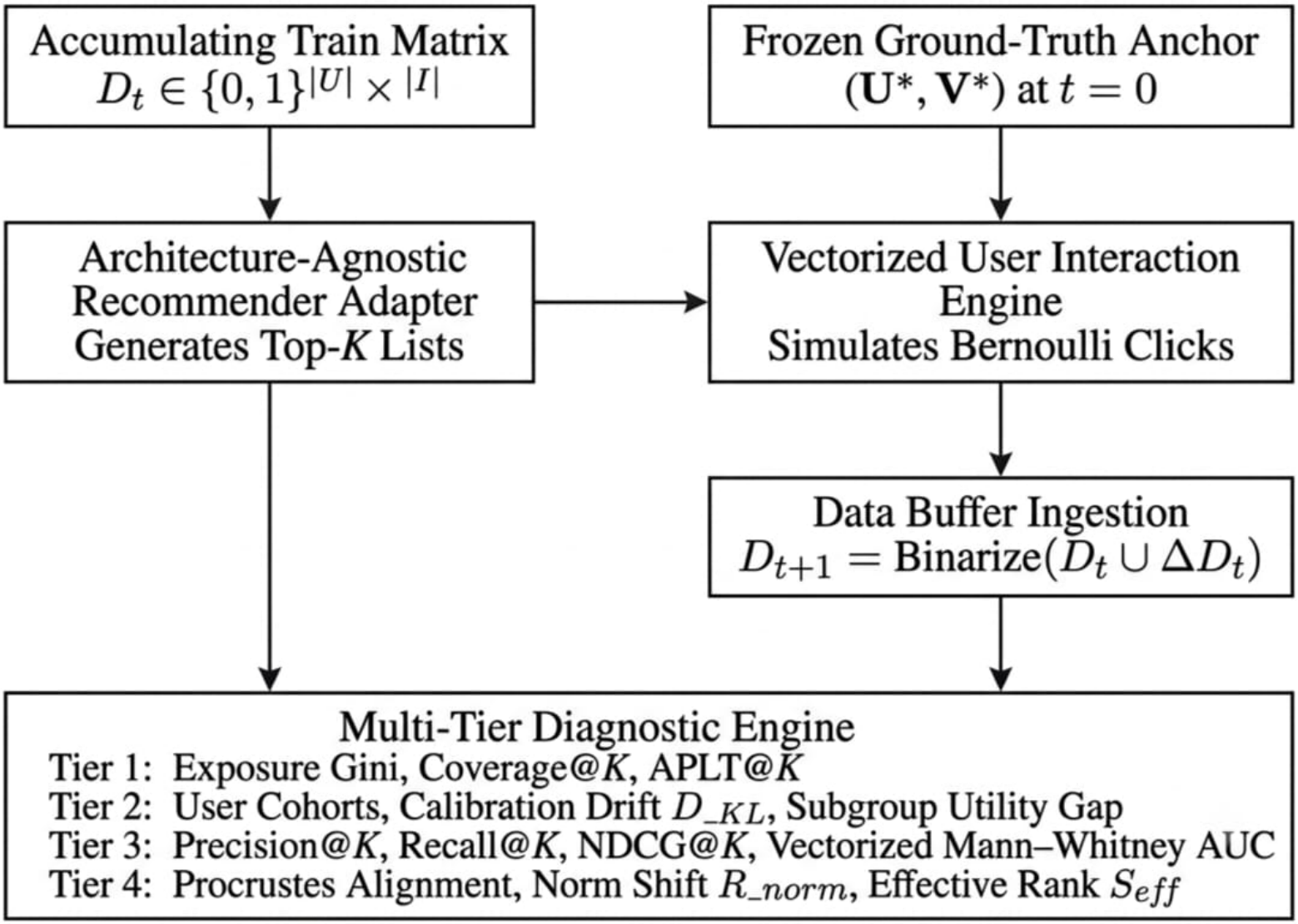


Figure: Proposed System Architecture

Flow Diagram of Proposed System

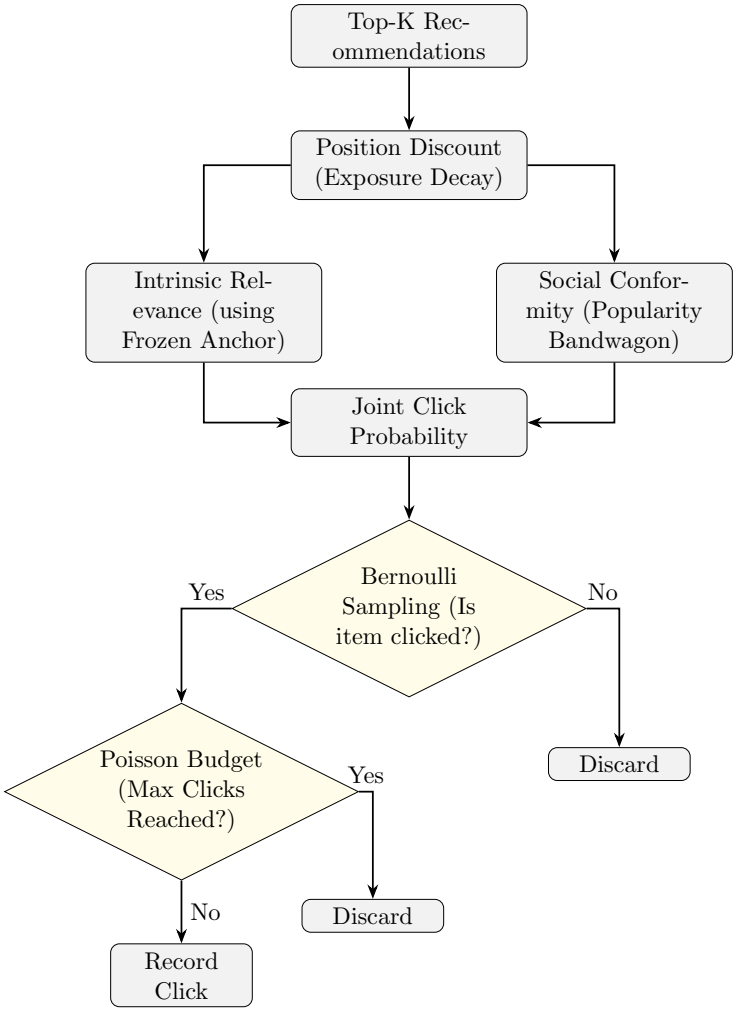


Figure: Vectorized User Interaction Flow

Block Diagram of Proposed System

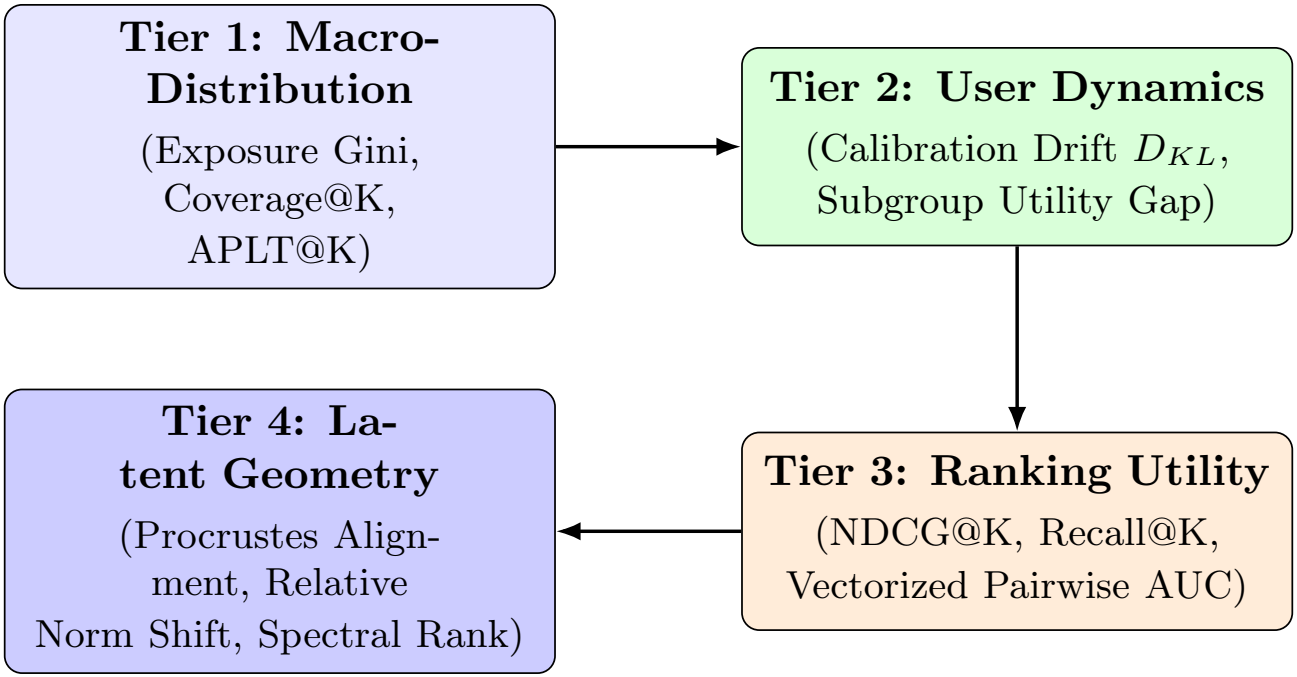


Figure: Multi-Tier Diagnostic Auditing Progression

Project Planning

Project Schedule

Phase	Activity	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Planning	Literature Survey	-						
	System Design	-	-					
Development	Data Pipeline & Preprocessing		-	-				
	Recommender Adapters (Cornac)			-	-			
	User Interaction Engine (Feedback Loop)				-	-		
	Diagnostic Auditing Engine (Metrics)					-	-	
Testing	Full-Loop Integration & Testing						-	-
	Performance Evaluation						-	-

Conclusion

- Proposed a robust closed-loop simulation framework to model popularity bias propagation in recommender systems.
- Integrates architecture-agnostic collaborative filtering models with a stochastic, vectorized user interaction engine.
- Enables comprehensive, multi-tier diagnostic auditing of macro-inequality, user taste distortion, and latent space degradation.
- Establishes a scalable baseline for evaluating bias mitigation algorithms and long-term representation fairness in continuous feedback cycles.

References I

[1] H. Zoralioglu, C. Yalcin, "Dynamic Feedback Loops in Recommender Systems," 2026.

[2] Y. Liu et al., "Causal Disentanglement for Tackling Popularity Bias," 2026.

[3] J. Chen, H. Dong, X. Wang, F. Feng, M. Wang, T.-S. Chua, "Bias and Debias in Recommender System: A Survey and Future Directions," *ACM Trans. Inf. Syst.*, vol. 41, no. 3, pp. 1–39, 2023.

[4] M. Mansoury, H. Abdollahpouri, M. Pechenizkiy, B. Mobasher, R. Burke, "Feedback Loop and Bias Amplification in Recommender Systems," in *Proc. 29th ACM Int. Conf. Inf. Knowl. Manag. (CIKM)*, 2020, pp. 2145–2148.

[5] A. Salah, Q. T. Truong, H. W. Lauw, "Cornac: A Comparative Framework for Multimodal Recommender Systems," *J. Mach. Learn. Res.*, vol. 21, no. 95, pp. 1–5, 2020.

References II

[6] J. Ma, C. Zhou, P. Cui, H. Yang, W. Zhu, "Learning Disentangled Representations for Recommendation," in *Proc. 33rd Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 5711–5722.

[7] A. J. B. Chaney, B. M. Stewart, B. E. Engelhardt, "How Algorithmic Confounding Increases Homogeneity and Decreases Utility," in *Proc. 12th ACM Conf. Recommender Syst. (RecSys)*, 2018, pp. 224–232.

[8] W.-C. Kang, J. McAuley, "Self-Attentive Sequential Recommendation," in *Proc. IEEE Int. Conf. Data Min. (ICDM)*, 2018, pp. 197–206.

[9] B. Hidasi, A. Karatzoglou, L. Baltrunas, D. Tikk, "Session-based Recommendations with Recurrent Neural Networks," in *Proc. 4th Int. Conf. Learn. Represent. (ICLR)*, 2016.

[10] S. Rendle, C. Freudenthaler, Z. Gantner, L. Schmidt-Thieme, "BPR: Bayesian Personalized Ranking from Implicit Feedback," in *Proc. 25th Conf. Uncertainty Artif. Intell. (UAI)*, 2009, pp. 452–461.

Thank You!